📘 2. REDOX REACTION – DETAILED SHORT NOTES

1. Oxidation & Reduction

Oxidation = loss of electrons

Reduction = gain of electrons

2. Oxidation Number (ON)

Rules:

Free element = 0

Oxygen = -2 (except peroxide = -1)

Hydrogen = +1 (metal hydride me -1)

3. Identifying Redox

ON change ho raha ho → redox

4. Balancing Redox (IMPORTANT)

Ion-electron method:

Step:

Oxidation and reduction half alag karo

Electrons balance karo

Charge aur atoms balance karo

Combine

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5. Oxidising vs Reducing Agent

Oxidising agent → khud reduce hota hai

Reducing agent → khud oxidise hota hai

📘 3. THERMODYNAMICS (PHYSICAL CHEMISTRY)

1. System & Surroundings

System = jo study kar rahe

Surroundings = baaki sab

2. Types of System

Open → mass + energy exchange

Closed → only energy

Isolated → no exchange

3. Internal Energy

$\Delta U = q - w$

4. Work Done

$w = -P\Delta V$

5. Enthalpy

$H = U + PV$

$\Delta H = q_p$ (constant pressure)

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6. Types of Processes

Isothermal → T constant

Adiabatic → no heat exchange

Isochoric → volume constant

7. Spontaneity

$\Delta G = \Delta H - T\Delta S$

Condition:

$\Delta G < 0$ → spontaneous

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